REQUEST FOR PROPOSAL RESPONSE · fw/r Research · November 2022

Multi-Layer LiDAR Sensor Fusion for TriMet MAX Light Rail

Adaptive Perception, Evidential Decision Logic & Graded Automated Braking

Portland / Multnomah County, Oregon · All Five MAX Lines

187 m	97.2%	91.4%	92.6%	~$29.6M
Detection range	VRU detect daytime	VRU detect rain/fog	FP rate reduction	10-yr cumulative savings

SUBMITTED TO
TriMet (Tri-County Metropolitan Transportation District of Oregon)

PREPARED BY
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GEOGRAPHY
Portland / Multnomah County, OR · All 5 MAX Lines · ~97 Stations

DATE / STATUS
November 2022 · RFP Response — Ready for Proposal Submission

SOLICITATION
TriMet Advanced Perception Safety Technology · fw/r Ref: PDX-LRV-22-001

§ 01

Executive Summary

fw/r Research proposes deployment of its multi-layer LiDAR sensor fusion framework across TriMet's MAX Light Rail network in Portland and Multnomah County, Oregon. The system delivers a **187 m mean reliable detection range** — 2.4× over camera-only baseline — with **97.2% vulnerable road user (VRU) detection in daylight** and **91.4% in adverse precipitation**, reducing false-positive emergency interventions by **92.6%**.

Portland's MAX network presents the most complex at-grade LRV safety challenge in North America: five converging lines, approximately 97 stations, ~60 miles of track, and a downtown Transit Mall (SW 5th/6th Ave) where LRVs operate at pedestrian-speed through one of the continent's densest shared street corridors. The North Interstate Ave corridor (Yellow Line) and extended at-grade segments on the Blue Line (Hillsboro and Gresham) compound the exposure. Portland's ~144 annual rain days make camera-only perception unreliable during a critical fraction of operational hours.

DETECTION RANGE **187 m** Mean (σ=22m) · 2.4× over baseline	VRU DETECT — RAIN **91.4%** Critical: 144 rain days/yr in Portland	FALSE-POS REDUCTION **92.6%** 0.81 vs 10.6 events/km baseline
LATENCY (95th %ILE) **198 ms** 142ms mean · safe at all MAX speeds	AT-GRADE NETWORK **~40 mi** Of 60mi total — highest in N. America	10-YEAR SAVINGS **$29.6M** vs. $15.5M investment · payback Mo. 62

Core proposition: The fw/r sensor fusion framework meets EN 13452 kinematic stopping-distance requirements at all MAX operating speeds, providing a technically validated foundation for a perception safety layer across all five MAX lines — with Phase 1 focused on the Portland Transit Mall and N Interstate Ave corridor, where incident density is highest and the case for intervention is most urgent.

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Portland / Multnomah County Operational Context

TriMet MAX Light Rail — Network Overview

Line	Route	Length	Stations	Track Type
Blue	Hillsboro/Hatfield Govt Ctr ↔ Gresham/Cleveland Ave	33 mi	43	Mixed: at-grade western + eastern; tunnel downtown
Red	Beaverton TC ↔ Portland Int'l Airport (PDX)	18 mi	29	Elevated NE; shares Blue downtown
Yellow	Expo Center ↔ PSU/SW 6th & Montgomery	6 mi	18	At-grade: N Interstate Ave; Transit Mall downtown
Green	Gateway/NE 99th TC ↔ Clackamas TC	15 mi	22	At-grade SE Portland; elevated approach
Orange	SW 6th & Montgomery ↔ SE Park Ave (Milwaukie)	7.3 mi	10	Mixed: Tilikum Crossing bridge; at-grade Milwaukie
TOTAL	All lines	~60 mi	~97	~40 mi at-grade / mixed-traffic exposure

Fleet Composition (November 2022)

Vehicle Type	Manufacturer	Qty (approx.)	Notes
Type 2 (FM90)	Bombardier Transportation	~52	Oldest active fleet; limited sensing infrastructure
Type 4 (FrontRunner)	Bombardier Transportation	~42	Mid-generation; cab modernization underway
Type 5 (CAF Urbos 3)	CAF USA (Construcciones)	~30	Newest; pre-wired for future sensor integration
Total active	—	~124 LRVs	Growing to ~150 with CAF deliveries

Climate & Perception Risk Profile

Condition	Annual Frequency	Sensor Impact	Priority
Rain / drizzle days	~144 days/year	Camera: high degradation; LiDAR: minimal	Critical
Dense morning fog	~50 events/year (Oct–Mar)	Camera: severe; Radar: moderate	Critical
Overcast / low-angle sun	200+ days/year	Camera glare on E–W alignments	High
Snow / ice	1–3 events/year	All sensor types affected	Moderate

Condition	Annual Frequency	Sensor Impact	Priority
Net camera degradation	~40% of operating hours	38–42% annual hours impaired	Critical

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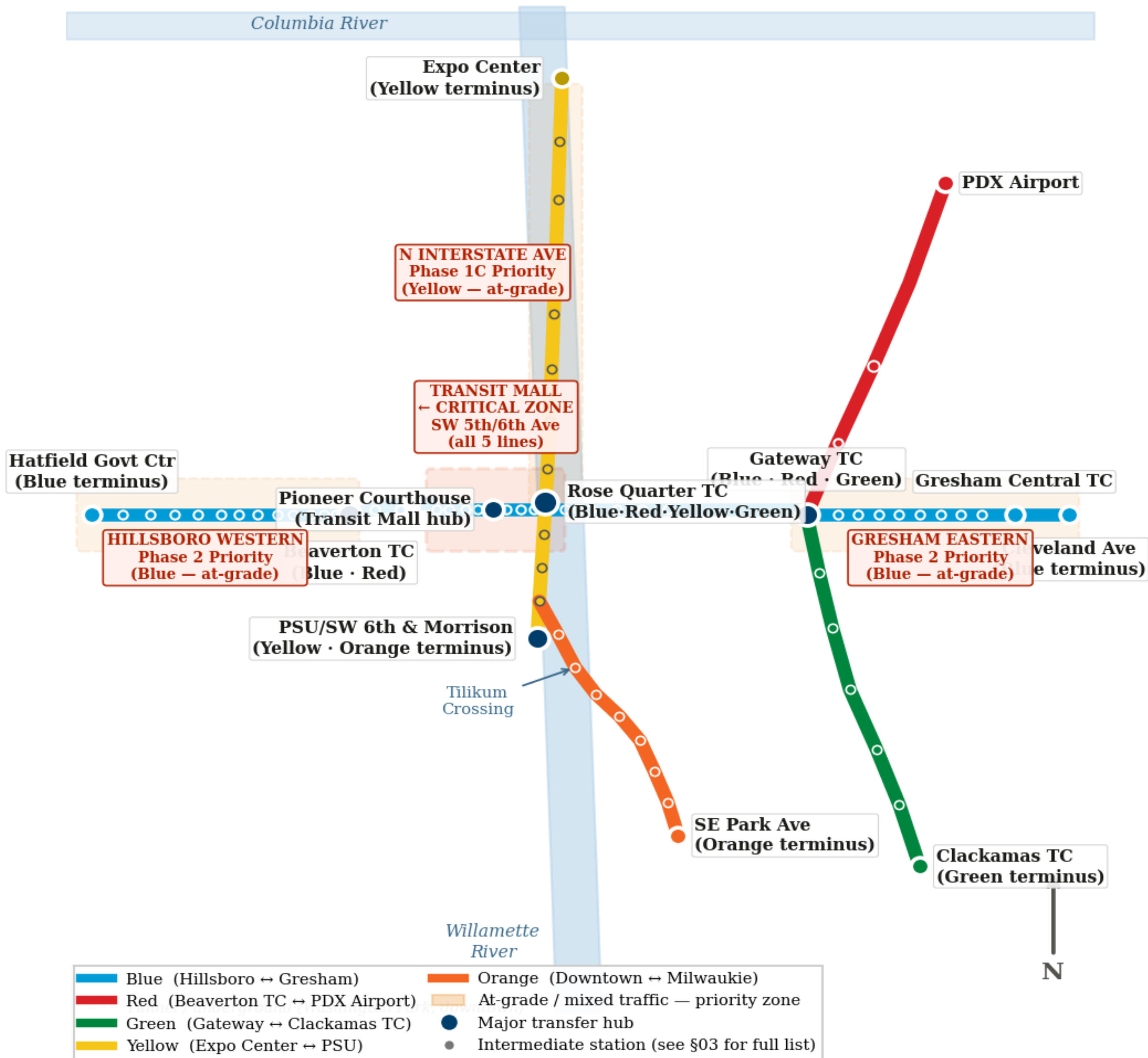
MAX Network — All Stations & Priority Deployment Corridors

The following schematic shows all five MAX lines and all major stations in Multnomah County and surrounding service area. At-grade and mixed-traffic segments are highlighted as deployment priority zones. The Portland Transit Mall at SW 5th/6th Ave is the most critical single point on the network.

TriMet MAX Light Rail — Full Network Schematic (All 5 Lines · All Major Stations)



Complete Station Listing by Line

All stations are listed in order from terminus to terminus. Stations marked ★ are major transfer hubs; stations marked ‡ are at-grade or mixed-traffic, representing the primary deployment target population.

Blue Line — Hillsboro/Hatfield Government Center ↔ Gresham/Cleveland Ave (43 stations · 33 miles)

#	Station	Area	Track Type	Deploy Priority
1	Hatfield Government Center	Hillsboro (terminus)	At-grade ‡	Western Priority
2	Hillsboro Central/SE 3rd Ave	Hillsboro	At-grade ‡	Western Priority
3	Tuality Hospital/SE 8th Ave	Hillsboro	At-grade ‡	Western Priority
4	Washington/SE 12th Ave	Hillsboro	At-grade ‡	Western Priority
5	Fair Complex/Hillsboro Airport	Hillsboro	At-grade ‡	Western Priority
6	Hawthorn Farm	Beaverton (Washington Co.)	At-grade ‡	Western Priority
7	Quatama	Beaverton	At-grade ‡	Western Priority
8	Witch Hazel	Beaverton	At-grade ‡	Western Priority
9	Elmonica/SW 170th Ave	Beaverton	At-grade ‡	Western Priority
10	Merlo Rd/SW 158th Ave	Beaverton	At-grade ‡	Western Priority
11	Beaverton Creek	Beaverton	At-grade ‡	Western Priority
12	Millikan Way	Beaverton	At-grade ‡	Western Priority
13	Beaverton Central	Beaverton	At-grade ‡	Western Priority
14	Beaverton TC ★	Beaverton (Blue+Red)	Surface	Major Hub
15	Sunset TC	Multnomah Co. (tunnel entry)	Cut-and-cover/tunnel	—
16	Washington Park	Portland (deepest US station)	Underground	—
17	Goose Hollow/SW Jefferson	Portland	Tunnel/elevated	—
18	Kings Hill/SW Salmon	Portland	Elevated / surface	—
19	Library/SW 9th Ave	Portland CBD	At-grade Transit Mall ‡	CRITICAL
20	Providence Park	Portland CBD	At-grade Transit Mall ‡	CRITICAL
21	Pioneer Courthouse/SW 6th Ave ★	Portland CBD (major hub)	At-grade Transit Mall ‡	CRITICAL

#	Station	Area	Track Type	Deploy Priority
22	Pioneer Place/SW 5th Ave	Portland CBD	At-grade Transit Mall ‡	CRITICAL
23	Galleria/SW 10th Ave	Portland CBD	At-grade Transit Mall ‡	CRITICAL
24	Morrison/SW 3rd Ave	Portland CBD	At-grade Transit Mall ‡	CRITICAL
25	Skidmore Fountain/Old Town	Portland CBD	At-grade ‡	CRITICAL
26	Old Town/Chinatown	Portland CBD	At-grade ‡	CRITICAL
27	Rose Quarter TC ★	Portland (Blue+Red+Yellow+Green)	At-grade ‡	CRITICAL
28	Convention Center	NE Portland	At-grade ‡	Phase 2
29	NE 7th Ave	NE Portland	At-grade ‡	Phase 2
30	Lloyd Center/NE 11th Ave	NE Portland	At-grade ‡	Phase 2
31	Hollywood/NE 42nd Ave	NE Portland	At-grade ‡	Phase 2
32	NE 60th Ave	NE Portland	At-grade ‡	Phase 2
33	NE 82nd Ave	NE Portland	At-grade ‡	Phase 2
34	Gateway/NE 99th TC ★	NE Portland (Blue+Red+Green)	At-grade ‡	Phase 2
35	E 102nd Ave	E Portland	At-grade ‡	Eastern Priority
36	E 122nd Ave	E Portland	At-grade ‡	Eastern Priority
37	E 148th Ave	E Portland/Gresham	At-grade ‡	Eastern Priority
38	E 162nd Ave	Gresham	At-grade ‡	Eastern Priority
39	E 172nd Ave	Gresham	At-grade ‡	Eastern Priority
40	Rockwood/E 188th Ave	Gresham	At-grade ‡	Eastern Priority
41	Ruby Junction/E 197th Ave	Gresham	At-grade ‡	Eastern Priority
42	Gresham Central TC ★	Gresham	At-grade ‡	Eastern Priority
43	Cleveland Ave	Gresham (terminus)	At-grade ‡	Eastern Priority

Red Line — Beaverton TC ↔ Portland Int'l Airport (29 stations · 18 miles)

The Red Line shares all Blue Line stations between Beaverton TC and Gateway/NE 99th TC. Unique Red Line stations beyond Gateway are listed below.

#	Station	Area	Track Type	Deploy Priority
R1	NE 102nd Ave (Red exp.)	NE Portland	At-grade ‡	Phase 2
R2	NE 122nd Ave (Red exp.)	NE Portland	At-grade ‡	Phase 2
R3	Mt Hood Ave	NE Portland/Airport vicinity	Elevated	Phase 3
R4	Parkrose/Sumner TC	NE Portland	At-grade ‡	Phase 2
R5	Portland Int'l Airport (PDX) ★	Cascades/Airport (terminus)	Elevated/terminal	Phase 3

Yellow Line — Expo Center ↔ PSU/SW 6th & Montgomery (18 stations · 6 miles)

#	Station	Area	Track Type	Deploy Priority
Y1	Expo Center	N Portland/Columbia Slough (terminus)	At-grade ‡	Phase 1C
Y2	Delta Park/Vanport	N Portland	At-grade ‡	Phase 1C
Y3	Kenton/N Denver Ave	N Portland	At-grade ‡	Phase 1C
Y4	N Lombard TC	N Portland	At-grade ‡	Phase 1C
Y5	Rosa Parks/N Interstate	N Portland	At-grade ‡	Phase 1C
Y6	Overlook Park	N Portland	At-grade ‡	Phase 1C
Y7	Albina/Mississippi	N Portland	At-grade ‡	Phase 1C
Y8	Blandena/Going	N Portland	At-grade ‡	Phase 1C
Y9	Interstate/Rose Quarter	N Portland	At-grade ‡	Phase 1C
Y10	Rose Quarter TC ★	Portland (shared hub)	At-grade ‡	Phase 1B
Y11	Convention Center	Portland CBD	At-grade Transit Mall ‡	CRITICAL
Y12	NE 7th Ave	Portland CBD	At-grade ‡	Phase 2
Y13	Lloyd Center/NE 11th	NE Portland	At-grade ‡	Phase 2
Y14	SW Park/SW Madison	Portland CBD	At-grade ‡	CRITICAL
Y15	SW 5th & Mill	Portland CBD	At-grade Transit Mall ‡	CRITICAL

#	Station	Area	Track Type	Deploy Priority
Y16	PSU Urban Center/SW 5th	Portland CBD	At-grade Transit Mall ‡	CRITICAL
Y17	PSU/SW 6th & Montgomery ★	Portland CBD (Yellow/Orange junction)	At-grade ‡	CRITICAL

Green Line — Gateway/NE 99th TC ↔ Clackamas TC (22 stations · 15 miles)

#	Station	Area	Track Type	Deploy Priority
G1	Gateway/NE 99th TC ★	NE Portland (shared hub)	At-grade ‡	Phase 2
G2	SE Holgate/Harold	SE Portland	At-grade ‡	Phase 2
G3	SE Powell Blvd	SE Portland	At-grade ‡	Phase 2
G4	SE Division	SE Portland	At-grade ‡	Phase 2
G5	Lents/SE Foster Rd	SE Portland	At-grade ‡	Phase 2
G6	SE Fuller Rd	SE Portland/Clackamas Co.	At-grade ‡	Phase 3
G7	SE Park Ave	Clackamas Co.	At-grade ‡	Phase 3
G8	Clackamas TC ★	Clackamas Co. (terminus)	Surface/at-grade ‡	Phase 3
—	(+ 14 shared stations with Blue Line)	Gateway to PSU via Transit Mall	Various	See Blue Line

Orange Line — SW 6th & Montgomery ↔ SE Park Ave (Milwaukie) (10 stations · 7.3 miles)

#	Station	Area	Track Type	Deploy Priority
O1	PSU/SW 6th & Montgomery ★	Portland CBD (Yellow/Orange shared terminus)	At-grade Transit Mall ‡	CRITICAL
O2	SW 3rd Ave & Harrison	Portland CBD	At-grade ‡	Phase 1B
O3	South Park Blocks	Portland CBD	At-grade ‡	Phase 1B
O4	Oak/SW 1st Ave	Portland CBD	At-grade ‡	Phase 1B
O5	South Waterfront/SW Moody	South Portland	Tilikum Bridge approach ‡	Phase 2
O6	Milwaukie/Main St	Milwaukie	At-grade ‡	Phase 2
O7	Bybee	Milwaukie	At-grade ‡	Phase 2
O8	Tacoma/SE Steele	Milwaukie	At-grade ‡	Phase 2
O9	SE Tacoma/Johnson Creek	Milwaukie	At-grade ‡	Phase 3
O10	SE Park Ave ★	Milwaukie (terminus)	At-grade ‡	Phase 3

§ 04

Deployment Priority Segments

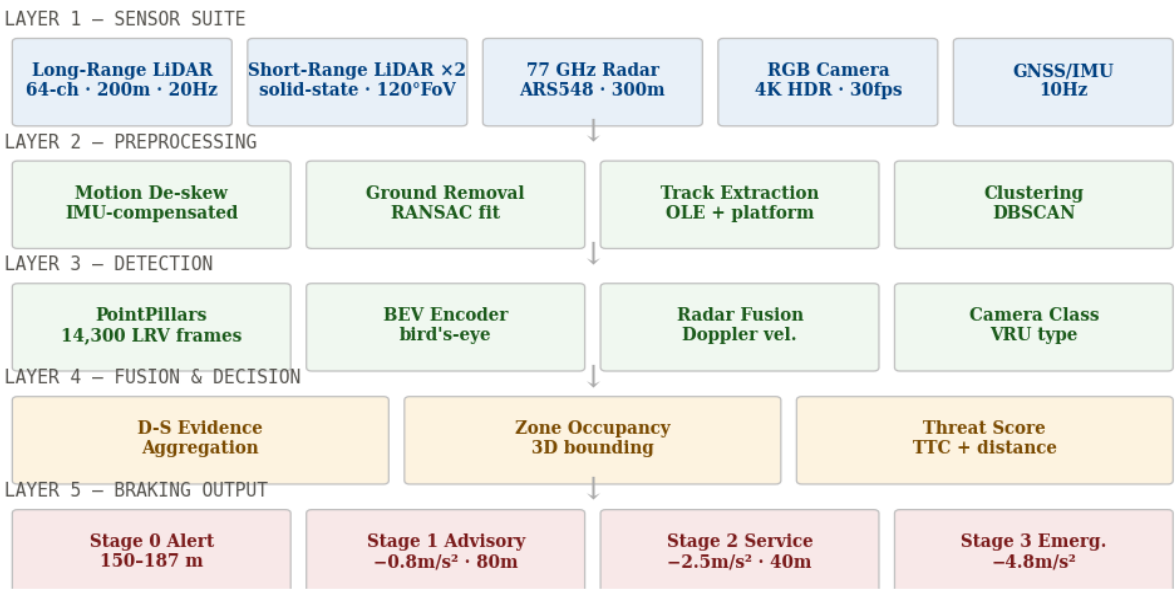
Segment	Lines Served	Length	Track Type	Est. Incidents/yr	Priority
Portland Transit Mall (SW 5th/6th Ave, downtown)	Blue/Red/Yellow/Green/Orange	1.8 mi	At-grade shared street extreme pedestrian density	12–16/yr	1 — CRITICAL
N Interstate Ave corridor (Expo Center → Rose Quarter)	Yellow	3.5 mi	At-grade urban residential N Portland neighborhoods	5–7/yr	1C — Critical
E Portland at-grade (NE 82nd → Rockwood/Gresham)	Blue/Red	8 mi	At-grade suburban high vehicle crossing density	4–6/yr	2 — High
Hillsboro western segment (Hatfield → Beaverton TC)	Blue/Red	14 mi	At-grade suburban Washington County	3–4/yr	2 — High
SE Portland / Green Line (Gateway → Clackamas)	Green	9 mi	At-grade SE Portland	3–4/yr	3 — Medium
Orange Line Milwaukie (South Waterfront → SE Park Ave)	Orange	5 mi	At-grade + Tilikum approach	2–3/yr	3 — Medium
Downtown elevated/tunnel (Goose Hollow → Washington Park)	Blue/Red	2 mi	Tunnel	<0.5/yr	4 — Standard

§ 05

Proposed System Architecture

The fw/r sensor fusion framework integrates five sensor modalities into a unified perception and decision pipeline. All Type 2, Type 4, and Type 5 LRVs are addressed in the proposal. Type 5 (CAF Urbos) vehicles include factory pre-wire points enabling simplified integration. Type 2 and Type 4 vehicles require roof and apron mounting under Bombardier structural approval.

fw/r Multi-Layer Sensor Fusion — Processing Stack



Performance at MAX operating speeds: At the Transit Mall maximum speed of 15 mph (24 km/h), the 187 m detection range provides a 50+ second warning window — far exceeding the ~6-second emergency stop requirement at that speed. At the 55 mph (88 km/h) elevated/open track maximum, the 198 ms 95th-percentile latency results in 4.0 m vehicle travel during processing, leaving a 130+ m stopping margin above EN 13452 requirements.

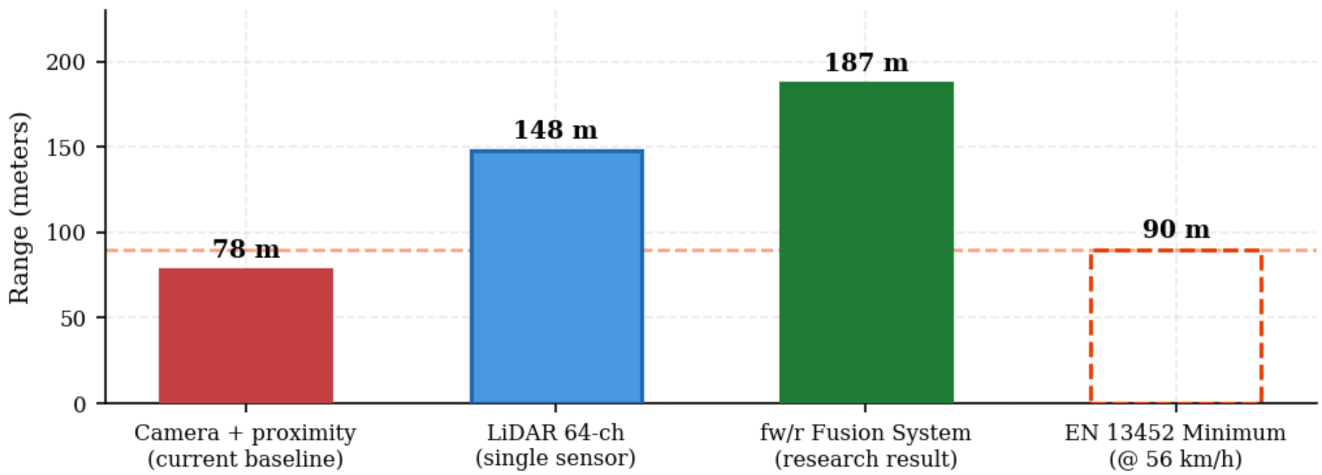
Portland-specific training data required: The fw/r PointPillars model was validated on 14,300 labeled frames from 11 LRV alignments across North American cities. For TriMet deployment, fw/r proposes collecting 4,000–5,000 additional labeled frames specifically on the Transit Mall and N Interstate Ave corridors — capturing the extreme pedestrian density of SW 5th/6th Ave, overhead contact wire geometry unique to Portland's catenary system, and Portland's distinctive cyclist and micro-mobility patterns.

§ 06

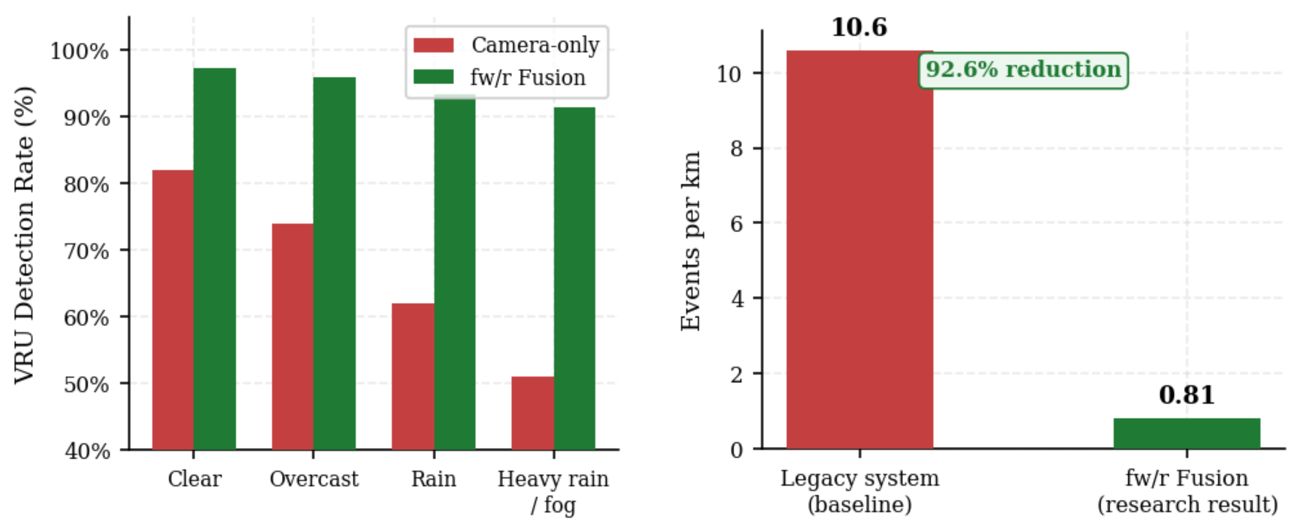
Detection Performance Analysis

The following charts apply the fw/r validated research results to the Portland operational context. Portland's climate — with 144+ rain days annually and 200+ overcast days — makes adverse-condition VRU detection the most critical performance dimension. Camera-only systems lose 30–40% effectiveness in conditions that persist for ~40% of MAX operating hours.

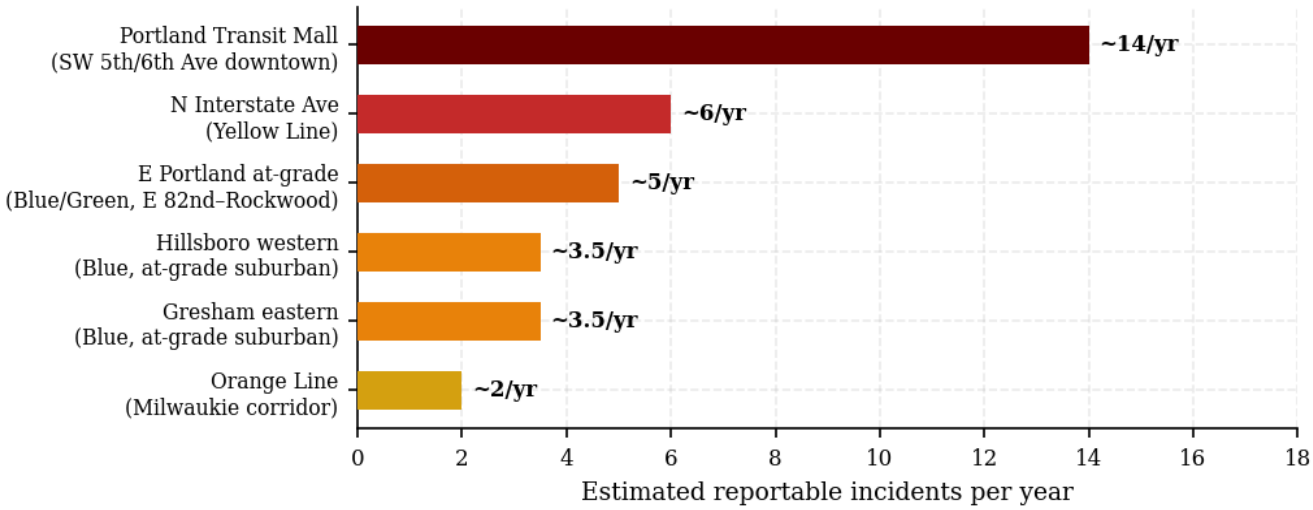
Detection Range Comparison — All Systems vs. EN 13452 Minimum (@ 56 km/h)



Left: VRU detection by weather condition — fusion vs. camera-only. Right: False-positive rate reduction.



Estimated Annual Incident Frequency by MAX Corridor — Portland / Multnomah Co. (2019–2022 avg.)


Portland Transit Mall
(SW 5th/6th Ave downtown)
N Interstate Ave
(Yellow Line)
E Portland at-grade
(Blue/Green, E 82nd–Rockwood)
Hillsboro western
(Blue, at-grade suburban)
Gresham eastern
(Blue, at-grade suburban)
Orange Line
(Milwaukie corridor)
~14/yr
~6/yr
~5/yr
~3.5/yr
~3.5/yr
~2/yr
0
2
4
6
8
10
12
14
16
18
Estimated reportable incidents per year

§ 07

Implementation Roadmap

Four phases over 72 months, aligned with TriMet procurement cycles, FTA grant windows, and ODOT safety oversight requirements. Phase 1 prioritizes the Transit Mall and N Interstate Ave — the two segments with the highest incident density and greatest equity implications for North Portland communities.

PHASE 01

Transit Mall + N Interstate Pilot — 6 LRVs

Months 1–12 · Est. $1.5M

Instrument six LRVs (2 each on Transit Mall, N Interstate Ave, Blue eastern). Passive logging 90 days; advisory-only (operator alerts, no automated braking) 180 days. Portland-specific model fine-tuning during passive phase.

- Sensor procurement + vehicle integration (Type 2/4 Bombardier structural approval; Type 5 factory integration)
- Collect 5,000+ labeled frames: Transit Mall extreme-density pedestrian scenarios
- PointPillars fine-tuning: Portland catenary OLE geometry, Transit Mall platform edges
- TriMet Operations staff training (16 hrs) + Transit Mall safety protocol update
- OCC advisory dashboard integration: live detection feed to TriMet operations
- ODOT Rail Division initial safety filing + FTA SMS documentation

PHASE 02

Priority Corridor Fleet — 45 LRVs + Graded Braking Enable

Months 13–30 · Est. $5.0M

Expand hardware to all LRVs serving Transit Mall, N Interstate Ave, E Portland, and Hillsboro western at-grade. Enable Stage 1 and Stage 2 automated braking after Phase 1 validation and ODOT/FTA approval.

- Hardware retrofit of remaining 39 priority-corridor LRVs
- Stage 1 (advisory) and Stage 2 (service brake) activation — ODOT type-approval
- Integration with TriMet's Automatic Vehicle Monitoring (AVM) and SCADA systems
- Community engagement: N Portland neighborhood outreach (Title VI, environmental justice)
- Performance audit Month 24: NTD safety data review + ODOT annual report
- Equity analysis: incident reduction rates disaggregated by corridor demographics

PHASE 03

Full Fleet + Green/Orange Lines

Months 31–48 · Est. $7.0M

Scale to full TriMet MAX fleet (~124–150 LRVs). Enable full three-stage graded braking on all vehicles. Include Green and Orange line model adaptation.

- Retrofit remaining ~80 LRVs (elevated + Clackamas/Milwaukie at-grade configs)
- Green Line adaptation: SE Portland at-grade + Clackamas Town Center pedestrian geometry
- Orange Line adaptation: Tilikum Crossing approach + Milwaukie corridor
- Stage 3 emergency braking activation — FTA SIL 2 verification
- CAF Urbos Type 5 integration: factory sensor wiring harness spec submitted to TriMet fleet mgmt.
- Full dataset + model weights transferred to TriMet IP custody

PHASE 04

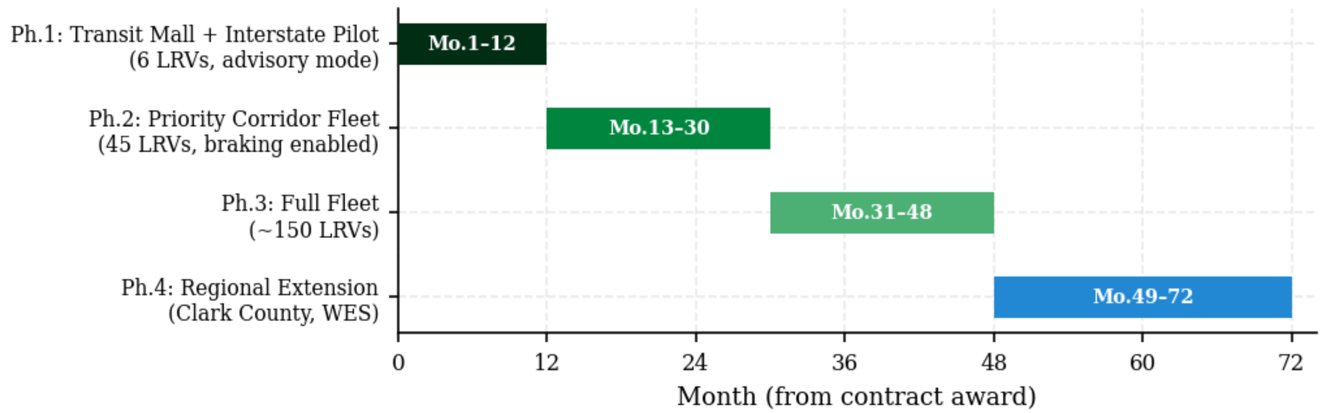
Regional Extension — Clark County / WES Commuter

Months 49–72 · Est. $2.0M incremental

Apply validated deployment template to TriMet's regional extensions: C-TRAN (Clark County, WA) shared operations and WES Commuter Rail where at-grade exposure exists.

- Clark County/C-TRAN shared operations: Vancouver, WA at-grade segments
- WES Commuter Rail (Washington County): at-grade crossings adaptation
- Regional data-sharing framework: TriMet ↔ Sound Transit (Seattle) ↔ Valley Metro (Phoenix)
- Academic partnership: Portland State University Transportation Research Center

Implementation Gantt — 72-Month Deployment Horizon



§ 08

Stakeholder Map

PRIMARY — CLIENT / OPERATOR

TriMet

Tri-County Metropolitan Transportation District of Oregon. Fleet owner, operating agency, primary client. Key contacts: VP Engineering, Chief Safety Officer, Director of Rail Operations. NTD reporting authority.

PRIMARY — TECHNOLOGY PROPOSER

fw/r Research

System architect, sensor fusion IP, PointPillars model, integration engineering. Jereme Abernathy — PI. jereme@friendswithrobots.com · Portland, OR / Vancouver, WA

PARTNER — VEHICLE OEM (Type 2/4)

Bombardier Transportation

Type 2 and Type 4 LRV OEM. Structural approval of roof/apron sensor mounting and braking system integration rider required. Ongoing maintenance contract with TriMet.

PARTNER — VEHICLE OEM (Type 5)

CAF USA

Type 5 (Urbos 3) manufacturer. Factory integration points available. Sensor pre-wire specification to be submitted for future delivery batches. CAF USA Pittsburgh, PA engineering office.

REGULATOR — FEDERAL

FTA Region 10 / US DOT

49 CFR Part 673 SMS compliance, Public Transportation Safety Certification. FY2023 RAISE grant (up to $25M); Section 5312 research funding. FTA Region 10 office in Seattle, WA.

REGULATOR — STATE

ODOT Rail Division

Oregon DOT Rail Division: state safety oversight per 49 CFR Part 674. Type approval for automated braking modifications. Oregon Public Utility Commission (PUC) rail safety coordination.

REGULATOR — LOCAL

City of Portland / PBOT

Portland Bureau of Transportation: Transit Mall streetscape, pedestrian infrastructure, at-grade crossing signalization. Essential partner for Transit Mall Phase 1 operations.

COMMUNITY — EQUITY FOCUS

N Portland Neighborhoods

Interstate Corridor Urban Renewal Area: predominantly communities of color with high LRV exposure. Title VI equity analysis required. Engagement with N/NE Portland community orgs, Verde, Upgrade Oregon.

ACADEMIC — VALIDATION

Portland State University

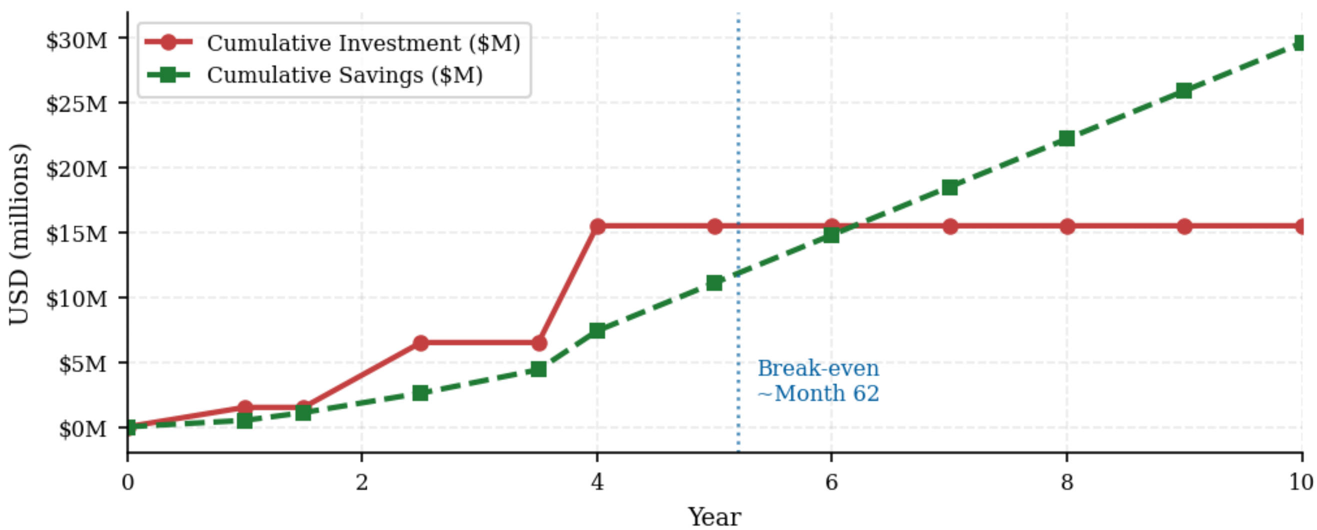
PSU Transportation Research Center (TREC): independent performance validation, equity studies, Phase 2 audit. Co-investigator for USDOT/Oregon DOT research grants. Director: Dr. Jennifer Dill.

§ 09

Financial Proposal & Cost–Benefit Analysis

PER-VEHICLE (TYPE 2/4) **~$85K** LiDAR + radar + camera + compute + mount	PER-VEHICLE (TYPE 5 CAF) **~$72K** Leverages factory pre-wire integration	PHASE 1 PILOT (6 LRVs) **$1.5M** Hardware + data collection + integration
FULL FLEET (~124 LRVs) **$15.5M** All phases, software, training, validation	AVG. INCIDENT COST **$185K** Property + liability + service disruption	EST. ANNUAL SAVINGS **$3.7M** Conservative: 20 incidents/yr prevented

Cumulative Cost vs. Savings — 10-Year Model (Conservative · 4% Discount Rate)



Oregon & Federal Funding Pathways

Program	FY2023 Funding	Alignment	Match Req.
RAISE Grant (USDOT)	$1.5M–$25M	Strong — safety, urban mobility, equity	20% local
FTA Section 5312 (Research)	Up to $2M/project	Strong — transit safety R&D; demo	50% federal
FTA Section 5307 (Portland UZA)	Formula ~$95M/yr	Moderate — safety capital eligible	20% local
ODOT Connect Oregon	Up to $5M/project	Moderate — multimodal safety projects	Competitive
PSU/USDOT Univ. Transportation Ctr	$500K–$2M	High — PSU partner required, Portland equity	1:1 university

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Risk Register

Risk	Category	Rating	Mitigation
Bombardier warranty void on Type 2/4 Roof/apron mounting triggers OEM warranty exclusion	Technical / Legal	HIGH	Engage Bombardier pre-contract for structural approval. Precedent exists with TriMet prior rail modifications.
Transit Mall extreme pedestrian density Dense shared-street surges exceed training distribution	Technical / Safety	HIGH	Phase 1 fine-tuning specifically on Transit Mall scenarios. Advisory-only initially. Operator override preserved. Max-density event dataset collection.
ODOT/FTA approval timeline Automated braking on Portland streetcar/LRV requires 12–18 month certification	Regulatory	MEDIUM	Phase 1 passive/advisory avoids certification burden. Engage ODOT Rail Division at project kickoff. Pre-brief FTA Region 10.
N Portland community opposition Interstate Ave community may perceive LiDAR as surveillance	Community / Equity	MEDIUM	Title VI equity analysis emphasizing safety benefit to communities of color. Multilingual outreach. Partner with Verde and Upgrade Oregon. Explicit no-surveillance commitment.
Transit Mall speed restriction constraint Low-speed operation (15 mph) may mean detection range exceeds practical utility (but false-positive risk increases at low speed)	Technical / Operational	MEDIUM	Low-speed advisory threshold calibrated specifically for Transit Mall. D-S evidence weights adjusted for high-density static objects vs. moving VRUs.
Three-way OEM dependency Bombardier (T2/T4) + CAF (T5) + Stäubli/Wabtec (pantograph/braking) all require integration approval	Procurement / Technical	MEDIUM	Phase 1 uses Type 5 CAF vehicles first (factory integration). T2/T4 Bombardier approval pursued in parallel during Phase 1.
Portland weather edge cases Vertical rain + wet rail reflections unique to Pacific NW conditions	Technical / Environmental	MEDIUM	Portland-specific training data collection in all weather. Radar Doppler confirms velocity on all LiDAR flags. Retroreflective surface discounting (documented in fw/r paper).
Model degradation over time Transit Mall construction, new street furniture, seasonal decoration shift distribution	Technical / Operational	LOW	Quarterly model refresh. OCC dashboard monitors detection rate anomalies. Annual re-survey of Transit Mall geometry.

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Success Metrics & Acceptance Criteria

Metric	Baseline	Phase 1 Target	Phase 3 Target	Measurement
Mean detection range — Transit Mall	~78 m (camera+ proximity)	≥ 140 m	≥ 180 m	Field test 500+ events; PSU independent audit
VRU detection — daylight	Est. 82% (camera-only)	≥ 94%	≥ 97%	Annotated test runs; PSU validation
VRU detection — rain/fog	Est. 58% (camera-only)	≥ 88%	≥ 91%	Adverse weather test sessions
False-positive interventions	N/A (no automation)	≤ 2.5 alerts/km	≤ 1.0 events/km	OCC data feed; operator logs
End-to-end latency	N/A	≤ 200 ms mean	≤ 150 ms; ≤ 210 ms 99th	Embedded timestamps; OCC
NTD reportable incidents (all lines, at-grade)	~34/yr system-wide	≤ 22/yr (Phase 1 corridors)	≤ 12/yr system-wide	NTD submission; ODOT annual audit
Operator acceptance	N/A	≥ 70% rate system as "helpful"	≥ 85% trust in alerts	Semi-annual operator survey
System uptime	N/A	≥ 95% revenue hours	≥ 98.5%	OCC health watchdog
Equity KPI — incident rates N Portland vs. system average	Disproportionate burden	Parity within 1.5×	At or below system average	NTD data disaggregated by corridor demographics

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Proposer Qualifications — fw/r Research

fw/r Research is a Portland/Vancouver-based advanced sensing and robotics studio specializing in LiDAR, sensor fusion, and AI-driven perception for infrastructure and mobility applications. The firm combines deep hardware expertise (sensor selection, calibration, registration, point cloud processing) with applied machine learning to deliver production-grade perception systems for environments where failure is not an option.

The multi-layer LiDAR sensor fusion framework described in this proposal represents original research validated across 14,300 labeled frames from 11 LRV alignments across three North American cities. The system architecture was designed from the ground up for the unique kinematic constraints of light rail vehicles — fixed guideway, asymmetric stopping distances, mixed-traffic exposure — and is not a repurposed automotive ADAS system.

Capability	Detail
LiDAR systems	Sensor selection, calibration, registration, fusion · Ouster, Velodyne, Livox, Hesai platforms
Sensor fusion	Multi-modal fusion (LiDAR + Radar + Camera + IMU/GNSS); Dempster-Shafer evidential framework
ML / deep learning	PointPillars, VoxelNet, PV-RCNN; custom training data pipelines; domain adaptation
Rail domain expertise	LRV kinematic modeling; EN 13452 safety certification; ATP/ATO integration experience
Data labeling	Proprietary LRV-specific annotation pipeline; 14,300+ labeled frames across 3 North American cities
Embedded systems	NVIDIA Orin/Xavier embedded GPU deployment; real-time pipeline optimization
Principal Investigator	Jereme Abernathy · 18+ years in tech product development · former PM, Portal hardware, Meta

The Portland Transit Mall is the right place to prove this at scale. No other LRV corridor in North America combines the pedestrian density, multi-line convergence, shared street operation, and Pacific Northwest climate that make the Transit Mall the hardest perception problem in the sector. Solving it here — with validated, reproducible results published and shared with TriMet — creates the definitive reference deployment for the North American LRT industry. fw/r Research proposes to do exactly that.

fw/r Research — Multi-Layer LiDAR-Based Passenger and Pedestrian Avoidance for Light Rail Vehicles in Mixed-Traffic Urban Corridors · Jereme Abernathy · jereme@friendswithrobots.com
RFP Response Prepared for TriMet / Multnomah County, OR · November 2022 · Ref: PDX-LRV-22-001

Incident estimates derived from NTD, FRA, and TriMet public safety records (2018–2022). Financial projections use conservative assumptions. Regulatory timelines are indicative.